CS 556: Mathematics for Machine Learning Assignment Template

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Question1

Write the complete solution of the following linear system

$$\begin{aligned} x + 2y - z &= 1 \\ 3x + 5y + 2z &= 3 \\ 2x + y + 13z &= 2 \end{aligned}$$

Solution

Given equations:

$$\begin{aligned} x + 2y - z &= 1 \\ 3x + 5y + 2z &= 3 \\ 2x + y + 13z &= 2 \end{aligned}$$

Step 1: Write the augmented matrix
We can write the system of equations as an augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 3 & 5 & 2 & 3 \\ 2 & 1 & 13 & 2 \end{array}\right]$$

Step 2: Perform Gaussian elimination
We will use Gaussian elimination to reduce the matrix to row echelon form.
Row 2: $R_2 \to R_2 - 3R_1$
Row 3: $R_3 \to R_3 - 2R_1$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -1 & 5 & 0 \\ 0 & -3 & 15 & 0 \end{array}\right]$$

Step 2.1: Simplify row 2
We multiply row 2 by -1 to make the leading coefficient in row 2 positive:
$R_2 \to -R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & -5 & 0 \\ 0 & -3 & 15 & 0 \end{array}\right]$$

Step 2.2: Eliminate the y-term in row 3
Next, we add 3 times row 2 to row 3 to eliminate the y-term in row 3:
Row 3: $R_3 \to R_3 + 3R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Step 3: Back substitution
Now, we have the matrix in row echelon form:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

From this, we can write the system as:

$$\begin{aligned} x + 2y - z &= 1 \\ y - 5z &= 0 \\ 0 &= 0 \end{aligned}$$

From the second equation, we solve for y:

$$y = 5z$$

Substitute this into the first equation:

$$x + 2(5z) - z = 1$$

$$x + 10z - z = 1$$

$$x + 9z = 1$$

$$x = 1 - 9z$$

Step 4: General solution
Thus, the general solution to the system is:

$$\boxed{\begin{aligned} x &= 1 - 9z \\ y &= 5z \\ z &= z \end{aligned}}$$

This is the solution in parametric form, where z is a free parameter. The solution set is:

$$(x, y, z) = (1 - 9z, 5z, z) \quad \text{for any} \quad z \in \mathbb{R}.$$

Conclusion:

The system has infinitely many solutions, and the solution is expressed in terms of the free parameter z.

Question2

Find the rank of the following matrix:

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

Solution

Given Matrix:

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

Step 1: Convert to Row Echelon Form

We start with the given matrix:

$$\begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

Since the first element is zero, we swap R_1 and R_2:

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

Step 2: Make the First Column Below the Pivot Zero

Perform the row operation:

$$R_3 \leftarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & -5 & -8 & -3 \end{bmatrix}$$

Step 3: Make the Second Column Below the Pivot Zero

Perform the row operation:

$$R_3 \leftarrow R_3 + 5R_2$$

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 2 \end{bmatrix}$$

Step 4: Normalize the Third Row

Divide the third row by 2:

$$R_3 \leftarrow \frac{1}{2} R_3$$

$$\begin{bmatrix} 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Since all three rows are nonzero, the rank of the matrix is given by the number of nonzero rows.

Final Answer

$$\boxed{\text{Rank} = 3}$$

Question3

Construct a matrix A whose column space contains the vectors

$$\begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}$$

and whose null space contains the vector

$$\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$$

Solution

Constructing a matrix A whose column space contains the vectors

$$\mathbf{v_1} = \begin{pmatrix} 3 \\ 6 \\ 2 \end{pmatrix}, \quad \mathbf{v_2} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix}$$

and whose null space contains the vector

$$\mathbf{v_3} = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}.$$

Step 1: Column Space Condition We start by constructing the matrix A where the first two columns are the given vectors $\mathbf{v_1}$ and $\mathbf{v_2}$. Thus, the matrix looks like

$$A = \begin{pmatrix} 3 & 4 & a \\ 6 & 0 & b \\ 2 & 1 & c \end{pmatrix}.$$

Step 2: Null Space Condition The vector $\mathbf{v_3}$ must be in the null space of A, meaning that:

$$A \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

This gives us the following system of equations:

$$3(2) + 4(2) + a(1) = 0, \quad 6(2) + 0(2) + b(1) = 0, \quad 2(2) + 1(2) + c(1) = 0.$$

Step 3: Solving for a, b, and c Solving the system of equations:

$$6 + 8 + a = 0 \quad \Rightarrow \quad a = -14,$$

$$12 + b = 0 \quad \Rightarrow \quad b = -12,$$

$$4 + 2 + c = 0 \quad \Rightarrow \quad c = -6.$$

Step 4: Construct the Final Matrix Substituting the values of a, b, and c back into the matrix A, we get

$$A = \begin{pmatrix} 3 & 4 & -14 \\ 6 & 0 & -12 \\ 2 & 1 & -6 \end{pmatrix}.$$

Verification To verify that this matrix satisfies the conditions:

- The first two columns are the given vectors $\mathbf{v_1}$ and $\mathbf{v_2}$, so the column space condition is satisfied. - The null space condition is satisfied because

$$A \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Thus, the matrix A is

$$A = \begin{pmatrix} 3 & 4 & -14 \\ 6 & 0 & -12 \\ 2 & 1 & -6 \end{pmatrix}.$$

Question4

Compute the matrix-vector multiplication using:

1. Linear combination of columns.
2. Dot product of rows.

$$\begin{bmatrix} 2 & 1 & 3 \\ 7 & 1 & 0 \\ 3 & 5 & 9 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

Solution

(1) Linear Combination of Columns

Matrix-vector multiplication can be interpreted as a linear combination of the columns of A, weighted by the corresponding entries of x:

$$Ax = 2 \begin{bmatrix} 2 \\ 7 \\ 3 \end{bmatrix} + 4 \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} + 1 \begin{bmatrix} 3 \\ 0 \\ 9 \end{bmatrix}$$

Computing each term:

$$2 \begin{bmatrix} 2 \\ 7 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 14 \\ 6 \end{bmatrix}, \quad 4 \begin{bmatrix} 1 \\ 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 20 \end{bmatrix}, \quad 1 \begin{bmatrix} 3 \\ 0 \\ 9 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 9 \end{bmatrix}$$

Adding them together:

$$Ax = \begin{bmatrix} 4 \\ 14 \\ 6 \end{bmatrix} + \begin{bmatrix} 4 \\ 4 \\ 20 \end{bmatrix} + \begin{bmatrix} 3 \\ 0 \\ 9 \end{bmatrix} = \begin{bmatrix} 11 \\ 18 \\ 35 \end{bmatrix}$$

(2) Dot Product of Rows

Alternatively, we compute each entry of Ax as the dot product of each row of A with x.

$$\begin{aligned} \text{First row:} \quad & (2,1,3) \cdot (2,4,1) = 2(2) + 1(4) + 3(1) = 4 + 4 + 3 = 11 \\ \text{Second row:} \quad & (7,1,0) \cdot (2,4,1) = 7(2) + 1(4) + 0(1) = 14 + 4 + 0 = 18 \\ \text{Third row:} \quad & (3,5,9) \cdot (2,4,1) = 3(2) + 5(4) + 9(1) = 6 + 20 + 9 = 35 \end{aligned}$$

Thus, the result is:

$$Ax = \begin{bmatrix} 11 \\ 18 \\ 35 \end{bmatrix}$$

Final Answer

$$\boxed{Ax = \begin{bmatrix} 11 \\ 18 \\ 35 \end{bmatrix}}$$

Question5

Find the value of k for which the matrix has:

1. Dependent columns
2. Independent columns

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 1 \\ 4 & 8 & k \end{bmatrix}$$

Solution

We are given the matrix:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 1 \\ 4 & 8 & k \end{bmatrix}$$

To determine the value of k for which the matrix has dependent or independent columns, we need to compute the determinant of the matrix.

Step 1: Compute the Determinant

The determinant of a 3×3 matrix is given by:

$$\det(A) = \begin{vmatrix} 1 & 3 & 2 \\ 2 & 3 & 1 \\ 4 & 8 & k \end{vmatrix}$$

Expanding along the first row:

$$\det(A) = 1 \begin{vmatrix} 3 & 1 \\ 8 & k \end{vmatrix} - 3 \begin{vmatrix} 2 & 1 \\ 4 & k \end{vmatrix} + 2 \begin{vmatrix} 2 & 3 \\ 4 & 8 \end{vmatrix}$$

We compute the 2×2 determinants:

$$\begin{vmatrix} 3 & 1 \\ 8 & k \end{vmatrix} = 3k - 8$$

$$\begin{vmatrix} 2 & 1 \\ 4 & k \end{vmatrix} = 2k - 4$$

$$\begin{vmatrix} 2 & 3 \\ 4 & 8 \end{vmatrix} = (2 \times 8) - (3 \times 4) = 16 - 12 = 4$$

Substituting these values back into the determinant formula:

$$\det(A) = 1(3k - 8) - 3(2k - 4) + 2(4)$$

Simplifying:

$$\det(A) = 3k - 8 - 6k + 12 + 8$$

$$\det(A) = -3k + 12$$

$$\det(A) = 12 - 3k$$

Step 2: Solve for Dependent Columns ($\det(A) = 0$)
For the columns to be dependent, the determinant must be zero:

$$12 - 3k = 0$$

Solving for k:

$$3k = 12$$

$$k = 4$$

Step 3: Solve for Independent Columns ($\det(A) \neq 0$)
For the columns to be independent, the determinant must be nonzero:

$$12 - 3k \neq 0$$

$$3k \neq 12$$

$$k \neq 4$$

Final Answer

(1) The columns are dependent if $k = 4$.
(2) The columns are independent if $k \neq 4$.

Question6

Find a basis for the four fundamental subspaces of the matrix A:

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

Solution

We are given the matrix:

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

The reduced row echelon form (RREF) of A is:

$$\text{RREF}(A) = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1. Column Space $C(A)$

The pivot columns in the RREF correspond to the 2nd and 4th columns of the original matrix. Therefore, the basis for $C(A)$ is:

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} \right\}$$

2. Row Space $C(A^T)$

The non-zero rows of the RREF form a basis for the row space:

$$\left\{ \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 2 \end{bmatrix} \right\}$$

3. Null Space $N(A)$

Since the rank of A is 2 and there are 5 columns, the null space has dimension $5 - 2 = 3$.

The basis for $N(A)$ is:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

4. Left Null Space $N(A^T)$

The left null space is the orthogonal complement of the row space of A. Since A has 3 rows and rank 2, its left null space has dimension $3 - 2 = 1$.

The basis for $N(A^T)$ is:

$$\left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Question7

Consider the vectors:

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

(a) In the x-y plane, mark all nine linear combinations $c\mathbf{v} + d\mathbf{w}$, where:

$$c = \{-2, 0, 2\}, \quad d = \{0, 1, 2\}$$

(b) What shape do all the linear combinations $c\mathbf{v} + d\mathbf{w}$ fill? A line? The whole plane? Are the vectors $\mathbf{v}$ and $\mathbf{w}$ independent?

Solution

Consider the vectors:

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

We are asked to find the nine linear combinations $c\mathbf{v} + d\mathbf{w}$, where:

$$c = \{-2, 0, 2\}, \quad d = \{0, 1, 2\}$$

The linear combinations $c\mathbf{v} + d\mathbf{w}$ are given by:

$$c\mathbf{v} + d\mathbf{w} = c \begin{bmatrix} 1 \\ 2 \end{bmatrix} + d \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} c + d \\ 2c \end{bmatrix}$$

Now, let's compute the nine linear combinations for the different values of c and d.

When $c = -2$:

- For $d = 0$, $c\mathbf{v} + d\mathbf{w} = \begin{bmatrix} -2 + 0 \\ 2(-2) \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$
- For $d = 1$, $c\mathbf{v} + d\mathbf{w} = \begin{bmatrix} -2 + 1 \\ 2(-2) \end{bmatrix} = \begin{bmatrix} -1 \\ -4 \end{bmatrix}$
- For $d = 2$, $c\mathbf{v} + d\mathbf{w} = \begin{bmatrix} -2 + 2 \\ 2(-2) \end{bmatrix} = \begin{bmatrix} 0 \\ -4 \end{bmatrix}$

When $c = 0$:

- For $d = 0$, $c\mathbf{v} + d\mathbf{w} = \begin{bmatrix} 0 + 0 \\ 2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- For $d = 1$, $c\mathbf{v} + d\mathbf{w} = \begin{bmatrix} 0 + 1 \\ 2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
- For $d = 2$, $c\mathbf{v} + d\mathbf{w} = \begin{bmatrix} 0 + 2 \\ 2(0) \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

When $c = 2$:

- For $d = 0$, $c\mathbf{v} + d\mathbf{w} = \begin{bmatrix} 2+0 \\ 2(2) \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$
- For $d = 1$, $c\mathbf{v} + d\mathbf{w} = \begin{bmatrix} 2+1 \\ 2(2) \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$
- For $d = 2$, $c\mathbf{v} + d\mathbf{w} = \begin{bmatrix} 2+2 \\ 2(2) \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$

Thus, the nine linear combinations are:

$$\begin{bmatrix} -2 \\ -4 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ -4 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ -4 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 4 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

These points form a grid in the x-y plane.

Part (b): Shape and Linear Independence

The linear combinations form a grid of points, not a single line, so they fill a rectangular grid in the plane. The two vectors $\mathbf{v}$ and $\mathbf{w}$ span the plane. Therefore, the set of all linear combinations does not form a line, but fills a grid within the plane.

Next, we check if the vectors $\mathbf{v}$ and $\mathbf{w}$ are linearly independent. To check this, we see if the equation

$$c\mathbf{v} + d\mathbf{w} = \mathbf{0}$$

has only the trivial solution. This leads to the system of equations:

$$c + d = 0$$

$$2c = 0$$

From the second equation, $c = 0$, and substituting into the first equation gives $d = 0$. Hence, the only solution is $c = 0$ and $d = 0$, which means that the vectors $\mathbf{v}$ and $\mathbf{w}$ are linearly independent.

Thus, the vectors $\mathbf{v}$ and $\mathbf{w}$ are **linearly independent**.

Question8

Consider the vectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

(a) Can you solve the system $x\mathbf{u} + y\mathbf{v} + z\mathbf{w} = \mathbf{b}$, if $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$?

(b) What if $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$? How many solutions are there?

(c) Are the vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ dependent or independent?

(d) Use parts (a)-(c) to decide if the matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

is invertible or not.

Solution

Consider the vectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

(a) Can you solve the system $x\mathbf{u} + y\mathbf{v} + z\mathbf{w} = \mathbf{b}$, if $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$?

We need to solve the system:

$$x \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + z \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

This corresponds to the following system of equations:

$$\begin{aligned} x + 2z &= 0 \quad (1) \\ -y + z &= 0 \quad (2) \\ x + y &= 1 \quad (3) \end{aligned}$$

From equation (2), we get $y = z$. Substituting this into equation (3):

$$x + z = 1 \quad \Rightarrow \quad x = 1 - z$$

Substitute $x = 1 - z$ into equation (1):

$$(1 - z) + 2z = 0 \quad \Rightarrow \quad 1 + z = 0 \quad \Rightarrow \quad z = -1$$

Now substitute $z = -1$ into $x = 1 - z$ and $y = z$:

$$x = 2, \quad y = -1, \quad z = -1$$

Thus, the solution to the system is:

$$x = 2, \quad y = -1, \quad z = -1$$

(b) What if $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$? How many solutions are there?

If $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, the system becomes:

$$x \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + z \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives the system of equations:

$$\begin{aligned} x + 2z &= 0 \quad (1) \\ -y + z &= 0 \quad (2) \\ x + y &= 0 \quad (3) \end{aligned}$$

From equation (2), we get $y = z$. Substituting this into equation (3):

$$x + z = 0 \quad \Rightarrow \quad x = -z$$

Substitute $x = -z$ into equation (1):

$$(-z) + 2z = 0 \quad \Rightarrow \quad z = 0$$

Thus, $x = 0$ and $y = z = 0$. The only solution to this system is the trivial solution:

$$x = 0, \quad y = 0, \quad z = 0$$

So, the system has only the trivial solution $(x, y, z) = (0, 0, 0)$.

(c) Are the vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ dependent or independent?

To check if the vectors are linearly dependent or independent, we form the matrix with $\mathbf{u}, \mathbf{v}, \mathbf{w}$ as columns and perform row reduction:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Performing row reduction:

1. Subtract row 1 from row 3 to eliminate the 1 in the first column of row 3:

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 1 & -2 \end{bmatrix}$$

2. Add row 2 to row 3 to eliminate the 1 in the second column of row 3:

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{bmatrix}$$

Since there are 3 pivot positions, the matrix has full rank (rank = 3), and therefore the vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent.

(d) Is the matrix $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ invertible?

Since the matrix has full rank (rank = 3), it is invertible.

Final Answers

(a) The solution is $x = 2$, $y = -1$, $z = -1$.

(b) The system has only the trivial solution $(x, y, z) = (0, 0, 0)$.

(c) The vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are linearly independent.

(d) The matrix A is invertible.

Question9

Consider the following linear system for some constants b and g:

$$x - 2y + 3z = 3$$
$$2x + y + bz = -4$$
$$x + 0y + z = g$$

(a) What constant b makes the system singular (missing a pivot)?

(b) For the value of b found in part (a), for which values of g does the system have infinitely many solutions?

(c) Find two distinct solutions of the system for that value of g.

Solution

Consider the following system of equations:

$$x - 2y + 3z = 3 \quad (1)$$
$$2x + y + bz = -4 \quad (2)$$
$$x + 0y + z = g \quad (3)$$

The augmented matrix for this system is:

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 2 & 1 & b & -4 \\ 1 & 0 & 1 & g \end{array}\right]$$

(a) What constant b makes the system singular (missing a pivot)?

To make the system singular, the determinant of the coefficient matrix must be zero. The coefficient matrix is:

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 1 & b \\ 1 & 0 & 1 \end{bmatrix}$$

We calculate the determinant of A:

$$\det(A) = \begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & b \\ 1 & 0 & 1 \end{vmatrix}$$

Expanding along the first row:

$$\det(A) = 1 \cdot \begin{vmatrix} 1 & b \\ 0 & 1 \end{vmatrix} - (-2) \cdot \begin{vmatrix} 2 & b \\ 1 & 1 \end{vmatrix} + 3 \cdot \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix}$$

Now, calculate the 2x2 determinants:

$$\begin{vmatrix} 1 & b \\ 0 & 1 \end{vmatrix} = 1$$

$$\begin{vmatrix} 2 & b \\ 1 & 1 \end{vmatrix} = 2 - b$$

$$\begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} = -1$$

Substitute these into the determinant expression:

$$\det(A) = 1 \cdot 1 - (-2) \cdot (2 - b) + 3 \cdot (-1)$$
$$\det(A) = 1 + 2(2 - b) - 3$$
$$\det(A) = 1 + 4 - 2b - 3 = 2 - 2b$$

For the system to be singular, the determinant must be zero:

$$2 - 2b = 0 \quad \Rightarrow \quad b = 1$$

Thus, the system is singular when $b = 1$.

(b) For the value of $b = 1$, for which values of g does the system have infinitely many solutions?

For $b = 1$, the augmented matrix becomes:

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 2 & 1 & 1 & -4 \\ 1 & 0 & 1 & g \end{array}\right]$$

We perform Gaussian elimination:
1. Subtract 2 times row 1 from row 2:

$$R_2 \to R_2 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & -5 & -10 \\ 1 & 0 & 1 & g \end{array}\right]$$

2. Subtract row 1 from row 3:

$$R_3 \to R_3 - R_1$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & -5 & -10 \\ 0 & 2 & -2 & g-3 \end{array}\right]$$

3. Subtract $\frac{2}{5}$ times row 2 from row 3:

$$R_3 \to R_3 - \frac{2}{5}R_2$$

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & -5 & -10 \\ 0 & 0 & 0 & g-3+\frac{20}{5} \end{array}\right]$$

Simplify the third row:

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & -5 & -10 \\ 0 & 0 & 0 & g+1 \end{array}\right]$$

For the system to have infinitely many solutions, the third row must be consistent. This requires:

$$g + 1 = 0 \quad \Rightarrow \quad g = -1$$

Thus, for $b = 1$, the system has infinitely many solutions when $g = -1$.

(c) Find two distinct solutions of the system for $g = -1$.

When $g = -1$, the augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & -2 & 3 & 3 \\ 0 & 5 & -5 & -10 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

From row 2, we have:

$$5y - 5z = -10 \quad \Rightarrow \quad y - z = -2 \quad \Rightarrow \quad y = z - 2$$

Substitute $y = z - 2$ into row 1:

$$x - 2(z - 2) + 3z = 3$$
$$x - 2z + 4 + 3z = 3$$

$$x + z = -1 \quad \Rightarrow \quad x = -1 - z$$

Thus, the general solution is:

$$x = -1 - z, \quad y = z - 2, \quad z = z$$

Now, we can choose two distinct values for z:
1. For $z = 0$, we get $x = -1$, $y = -2$, and $z = 0$. So one solution is $(-1, -2, 0)$.
2. For $z = 1$, we get $x = -2$, $y = -1$, and $z = 1$. So another solution is $(-2, -1, 1)$.
Thus, two distinct solutions are:

$$(-1, -2, 0) \quad \text{and} \quad (-2, -1, 1)$$

Final Answers

(a) The system is singular when $b = 1$.

(b) The system has infinitely many solutions when $g = -1$ and $b = 1$.

(c) Two distinct solutions of the system for $g = -1$ are $(-1, -2, 0)$ and $(-2, -1, 1)$.